3.5.1 Wavelength at fixed wave speed

In the same medium under unchanged conditions, wave speed stays fixed, so wavelength is inversely proportional to frequency: doubling frequency halves wavelength. On crossing into a different medium, source frequency remains unchanged while speed and wavelength change together.

$$\lambda = \frac{v}{f}$$

At fixed wave speed v, increasing frequency f decreases wavelength λ. v: m s^{-1}; f: Hz; λ: m.

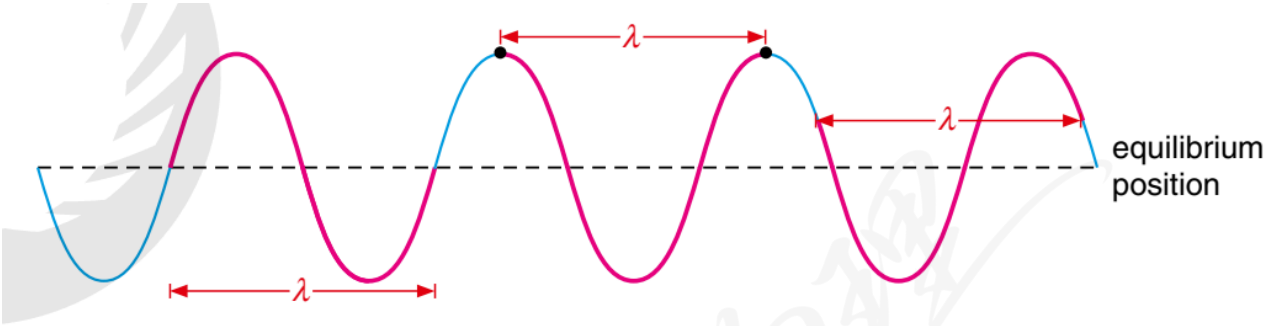


Fig 4.2b *Wavelength of a transverse wave.*

One wavelength separates adjacent points in phase, such as neighbouring crests, measured along propagation.

3.5.2 Reflection of waves

Reflection returns a wave from a boundary into its original medium. Incidence and reflection angles, measured from the normal, are equal. Since the medium is unchanged, speed, frequency and wavelength remain unchanged. This angular rule is separate from phase reversal, which depends on the boundary conditions, such as reflection at a fixed end.

$$r = i$$

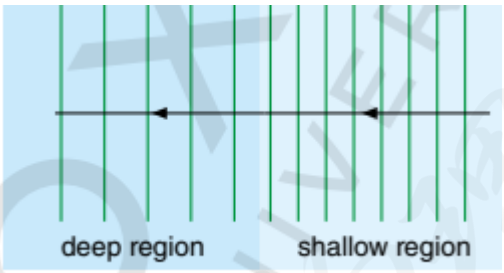
Reflection: incidence angle i and reflection angle r are measured from the normal to the reflecting surface.

3.5.3 Refraction of water waves

Water waves entering shallower water slow down while frequency stays constant, so wavelength decreases. Oblique rays bend towards the normal. The reverse occurs on entering deeper water. Refraction angles are related to speed or wavelength ratios and are measured from the normal. At normal incidence, speed and wavelength change even though direction does not.

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

Refraction: frequency remains unchanged. Angles are from the normal; v_1, v_2: wave speeds; λ_1, λ_2: wavelengths.



Wavelength increases as the wave enters deeper water. This normal-incidence example has no direction change.

3.5.4 Diffraction

Diffraction is the spreading of waves through a gap or around an obstacle into the geometrical shadow. Spreading becomes more pronounced with a longer wavelength at fixed gap width, or a narrower gap at fixed wavelength. It is readily visible when wavelength and gap width are comparable. Diffraction within one medium does not require a speed change.

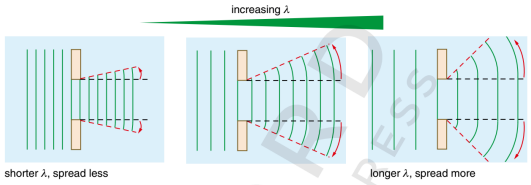


Fig 5.3c *Effect of λ on diffraction patterns.*

When the gap width decreases, the wave spreads more (Fig 5.3d).

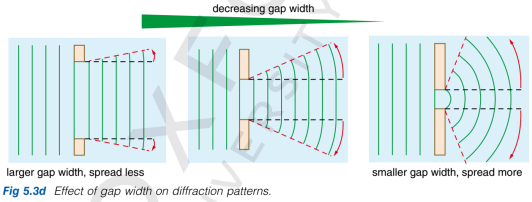
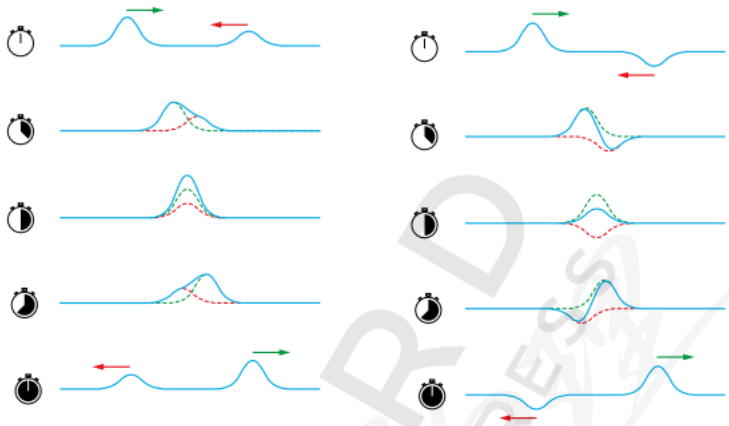


Fig 5.3d *Effect of gap width on diffraction patterns.*

Compare one variable at a time: greater wavelength at fixed gap, or a narrower gap at fixed wavelength, produces greater spreading.

3.5.5 Superposition

When waves overlap, their resultant displacement at each point is the signed sum of individual displacements. Add displacements, not energies directly. After meeting, pulses in an ideal linear medium continue in their original directions. Temporary cancellation does not mean the waves have been permanently destroyed.



Displacements add while the pulses overlap; each pulse then continues travelling.

3.5.6 Interference and path difference

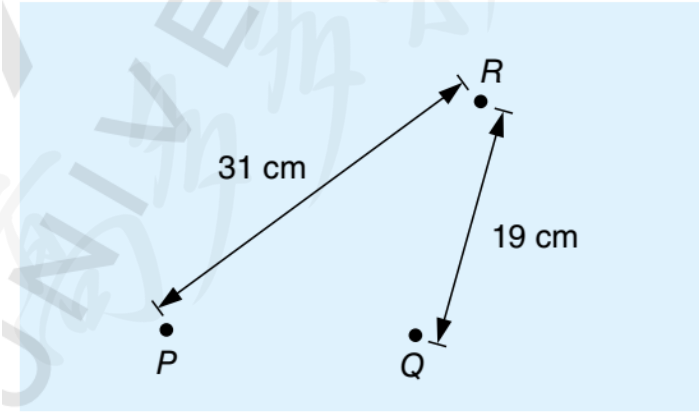
Waves of equal frequency and stable phase difference produce a stable interference pattern. For in-phase sources, integer-wavelength path differences give constructive interference and half-integer differences give destructive interference. Complete cancellation also requires equal amplitudes. Initially antiphase sources interchange these conditions, so source phase must be considered as well as path difference.

Spacing of two-source interference patterns

For coherent, in-phase water-wave sources in the same observation region, a longer wavelength or smaller source separation spreads neighbouring nodal and antinodal lines farther apart. The observation direction must then change through a larger angle to reach the next interference condition. Keep other conditions fixed when comparing patterns.

Evidence for wave behaviour

Reflection, refraction, diffraction and interference can all occur with waves. Diffraction and interference are particularly characteristic of wave spreading and superposition. Reflection or a deflected path alone does not establish a wave model.



For in-phase sources P and Q, the path difference at R is 31 − 19 = 12 cm. With wavelength 4 cm, this is three wavelengths and gives constructive interference.

3.5.7 Pulse reflection at a fixed end

A pulse reflected at a fixed end is inverted: crests become troughs and propagation reverses. In the same medium, wave speed stays unchanged, as do frequency and wavelength for periodic waves. When drawing the reflected pulse, reverse both its sign and its left-right profile: the part reaching the end first also leaves first, so a vertical flip alone is insufficient.

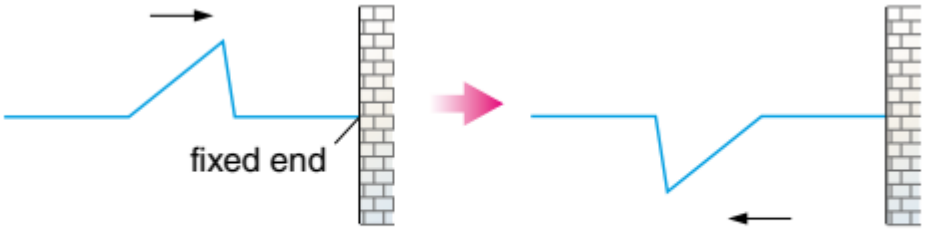


Fig 5.4c *The reflection of a pulse at a fixed end.*

Reflection at a fixed end inverts a crest into a trough and reverses propagation.

3.5.8 Stationary waves

Oppositely travelling waves with equal frequency, amplitude and speed can form fixed nodes and antinodes. Nodes have zero amplitude and antinodes maximum amplitude. Adjacent nodes are half a wavelength apart; a node and its nearest antinode are a quarter wavelength apart.

All particles away from nodes oscillate at the same frequency, but amplitude varies with position. Particles between two adjacent nodes are in phase and move in the same direction together. Neighbouring loops separated by a node are in antiphase: one moves upwards as the other moves downwards. Nodes remain at rest.

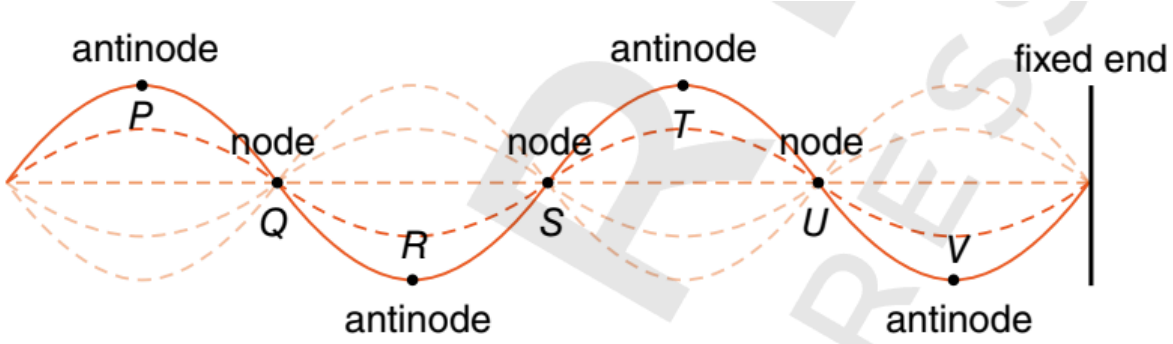


Fig 5.4f *Nodes and antinodes on a stationary wave.*

Nodes remain stationary; antinodes have maximum amplitude. Adjacent nodes are half a wavelength apart, and a node-to-adjacent-antinode gap is a quarter wavelength.